Practical 01

Exercise - I

a. Study of Network Hardware & Functionalities
b. Create a LAN Configuration using Hub and Switch

Procedure - Installation of CISCO Packet Tracer and study exercise:

- First, look up “Cisco Packet Tracer” or similar, on your search engine of choice.
- You should find a webpage on the Cisco Networking Academy site. Click on that, then click on “View Courses”.
- You should now have been redirected to the Skills For All website. Find the “Getting Started with Cisco Packet Tracer” course, and click on “Get Started”.
- Log in with any account of your choice.
- You should now be able to access the course material. On one of the first few pages, you should be able to find a link that redirects you to a Resources page.
- you can download (and, subsequently, install) the right version.

Contd…

- After looking through all of the various menus and options, trying out a few devices, testing how connections work etc.
- Select any two End Devices of your choice, and move them onto the main area.
- Select a switch, and a server, and move them to the main area as well.
- Connect them with wires as appropriate (there is an option titled "Automatically Choose Connection type" which picks the right one for you).
- A network with two end devices has now been created!

Basic network commands

Open the command prompt (cmd) from the Laptop and try the following basic commands – take the screenshot for each command

- color
- ipconfig
- ipconfig/all
- cls
- nslookup google.com
- control c
- Ping google.com

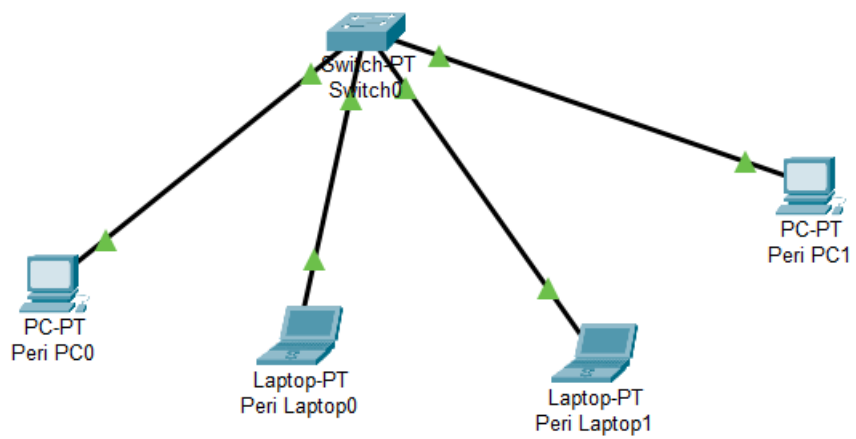
Procedure: Create a LAN Configuration using a Switch and Hub

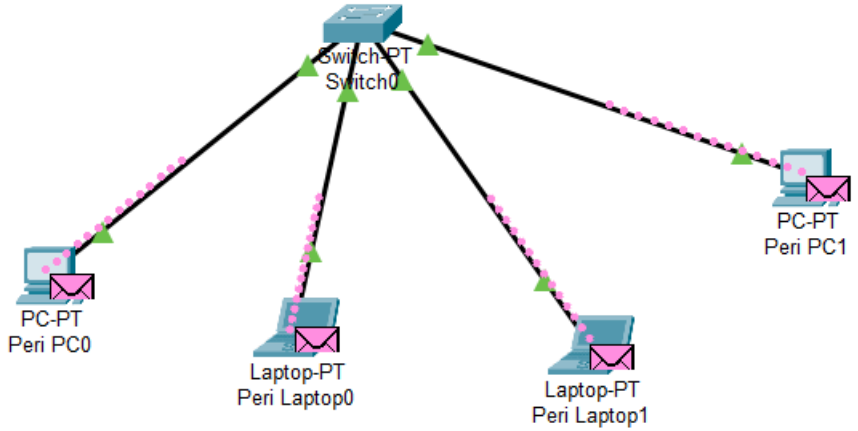
- Startup Cisco Packet Tracer, and place any 4 end devices of your choice.
- Connect them all using a switch/ Hub.
- For each of them: double-click the device, go to the Desktop tab, click on IP Configuration and set an IP.
- Make sure to give them all similar IPs – for eg, if an address is p.q.r.s, keep the ‘p’, ‘q’, and ‘r’ values (1 to 254) equal for all 4 end devices.

Contd…

- In one of the toolbars, on the top left, click on ‘Add Simple PDU’.
- Select one of the devices to be the sender of the message, and another to be the recipient.
- Click on ‘Simulation’ in the bottom right, and start the simulation of the Scenario.
- Take screenshots as and when necessary, throughout the duration of the simulation.

Sample - Screenshots





Sample – IP Configuration

